

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR10074 COMPUTER COMMUNICATIONS AND NETWORKS

Friday 17th May 2024

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. Answer QUESTION 1 and ONE other question.
2. Question 1 is COMPULSORY. If both QUESTION 2 and QUESTION 3 are answered, only QUESTION 2 will be marked.
3. All questions carry equal weight.
4. This is a NOTES PERMITTED, CALCULATORS NOT PERMITTED examination.

Candidates may consult up to THREE A4 pages (6 sides) of notes.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION.

Year 3 Courses

Convener: D.Armstrong

External Examiners: J.Bowles, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. THIS QUESTION IS COMPULSORY

- (a) Consider a host that does not have an IPv4 address yet but wishes to send an HTTP GET request to the HTTP server that resides in another broadcast domain. The network that the host resides in provides the following: a DNS server that knows the IP address of the HTTP server, a gateway router that provides connectivity to the HTTP server, and a DHCP server that provides the IP address of the host itself, DNS server and gateway to the host. The host implements IPv4, ARP, ICMP, DHCP, UDP, TCP and HTTP. List all the packets sent or received by the host before sending the packet that contains the HTTP GET message in the correct order, and provide a description of each packet (maximum 20 words each, and include source and destination *e.g.*, “ICMP Echo Request packet sent from the host to the gateway”). [11 marks]
- (b) Suppose a link that transfers approximately $1/p$ consecutive packets followed by a single packet drop. In a sender that employs the AIMD algorithm, the congestion window will traverse a periodic sawtooth, where the maximum and minimum window sizes will be W and $W/2$, respectively. We want to derive the additive increase parameter a that achieves 2 times higher throughput than regular TCP where a is the number of packets by which the window size increases every RTT (regular TCP thus uses $a = 1$). Note that the multiplicative decrease parameter is the same with TCP (i.e., $1/2$). Ignore link utilization of ack packets.
- Define the amount of data sent in a single cycle, where the window increases from $W/2$ until W , using W and a . [4 marks]
 - Define W using a and p . [4 marks]
 - Define the throughput T using a , p , the packet size S (assume all the packets have the same size), RTT R and W . [4 marks]
 - From the result obtained in 1(b)iii, derive a that achieves 2 times higher throughput of regular TCP. [2 marks]

2. ANSWER EITHER THIS QUESTION OR QUESTION 3

- (a) Suppose a host has a 4-MB file that is to be sent to another host over a 3.2-Mbps link with 10 ms one-way propagation delay. The file takes 1 second of CPU time to compress the data by 45%, or 2 seconds to compress by 50%. Ignore queueing delays, and compression and transmission cannot overlap. Calculate the total time of compression time, transmission delay and propagation delay for each of the compression options. [5 marks]
- (b) Consider a firewall that mediates every TCP packet sent or received by a host. This firewall shifts the value of the sequence number field in every TCP packet sent by the host by N , where N is the same value for the same TCP connection, and updates the checksum field in the TCP header. Can the host communicate with another host outside the firewall? Why? Note that there is no other packet modification made by this firewall. (your answer should be at most 40 words long) [5 marks]
- (c) Explain a possible reason why web caches would prevent the server and client from the use of new TCP extensions that rely on a new TCP option. (your answer should be at most 50 words long) [5 marks]
- (d) TCP Reno implements the fast retransmit mechanism, but there are still situations when the sender needs to wait for timeout expiration before retransmission even in case of a single packet loss. Explain when this situation happens. (your answer should be at most 60 words long) [3 marks]
- (e) Suppose a local network with a NAT gateway that has a single global IP address. How many TCP connections can exist at the same time for the same remote (outside the local network) address and port? [3 marks]
- (f) Explain two problems with AIMD (TCP Reno) over high bandwidth, long delay networks. (your answer should be at most 30 words long) [4 marks]

3. ANSWER EITHER THIS QUESTION OR QUESTION 2

- (a) Describe why OSPF is more scalable (to larger networks) than RIP? (your answer should be at most 80 words long) [3 marks]
- (b) Suppose a large organization with 1000 routers employs a simple 2-level routing hierarchy with the routers equally distributed between 20 areas (as in OSPF) to keep the routing table size small. Assume that, unlike OSPF, every router needs to keep an entry in its routing table indicating the route to reach areas other than the one it belongs to. At what network size in terms of the total number of routers will it no longer be possible to keep the routing table size at or below 100 entries with the above mentioned 2-level hierarchy? [10 marks]
- (c) Consider a four-port (labeled as 1-4) self-learning switch. Computer A and B connect to switch port 3 and 4, respectively, and switch port 1 and 2 are connected to each other by a cable. After computer A sends one packet to B, the network becomes almost unusable. Explain why this happens by referring to input and output port(s) of relevant packet(s) (your answer should be at most 60 words long). [4 marks]
- (d) What is the longest-possible common network prefix and its length of two IPv4 addresses, 192.168.55.1 and 192.168.58.2? [4 marks]
- (e) Consider the following IPv4 routing table based on longest prefix matching. We want X to be the network address with the 23-bit subnet mask such that packets whose destination is 133.27.171.224 are forwarded to interface 3. What is X?
- ```
0.0.0.0/0 interface 3
133.27.160.0/20 interface 1
133.27.186.0/24 interface 2
X/21 interface 3
```
- [4 marks]